Embedded System Design and Modeling

VI. Composition of State Machines

Composition of state machines

- How to combine multiple smaller FSMs into a bigger one?
- What problems arise?
- Two types of compositions:
  1. **Spatial** composition: how are the components connected?
  2. **Temporal** composition: how do the components react in time?

Spatial composition

Spatial composition = how are two components connected, how does the information flow between the components

- **Side-by-side** composition = no common inputs/outputs, no shared data
- **Cascade** composition = Outputs of one FSM are inputs to another one
- **Feedback** composition = (Some) outputs of an FSM are inputs to the same FSM, or to some other component situated upstream

Side-by-side composition

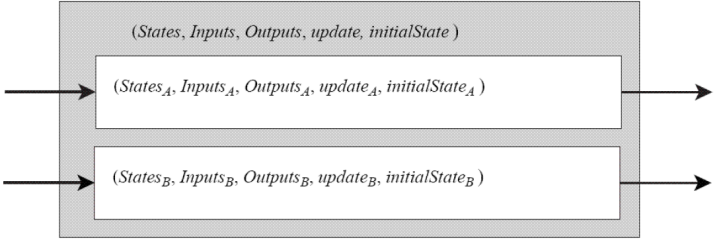


Figure 1: Side-by-side composition

Cascade composition

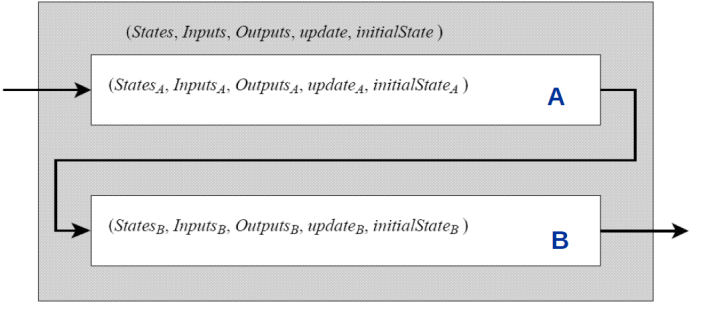


Figure 2: Cascade composition

- Outputs of FSM A are inputs to FSM B

Feedback composition

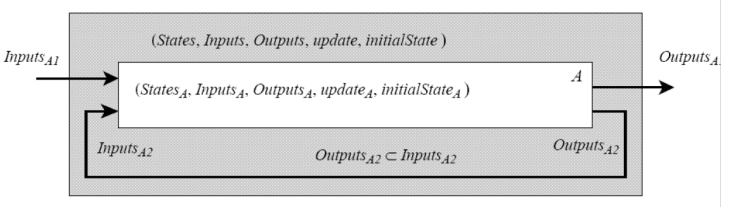


Figure 3: Feedback composition

- Some outputs of the FSM are coming back as inputs

Temporal composition

Temporal composition = when do two components react?

- **Sequential vs Parallel** composition:
  - **Sequential** = the two FSMs do not work at the same time
  - **Parallel** = the two FSMs work at the same time
- **Asynchronous vs Synchronous** composition = only for parallel composition
  - **Synchronous** = the FSMs react at the same time, i.e. are driven by the same clock
  - **Asynchronous** = the FSMs react at independent times

Sequential composition



Figure 4: Example of Sequential composition

- https://www.youtube.com/watch?v=iD3QgGpzzIM
- The drone has three modes of operation, working **in sequence**

Parallel composition

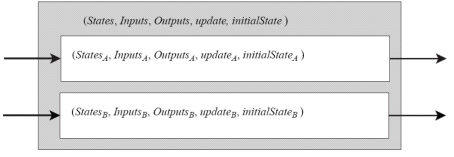


Figure 5: Side-by-side composition

- The two FSMs can be represented as a single **equivalent model** FSM
- When do the transitions in these FSMs take place?
  - Synchronous: simultaneously
  - Asynchronous: independently

Synchronous composition

- Consider the two FSMs on the left (A and B)
- The equivalent model is on the right

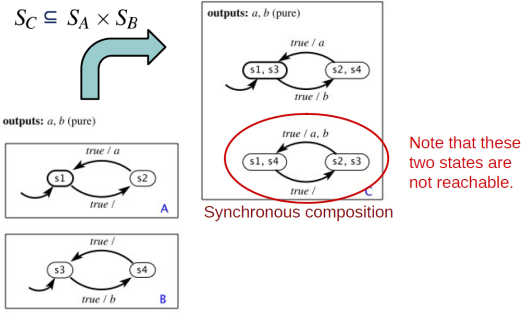


Figure 6: Synchronous composition

Synchronous composition

Redraw here

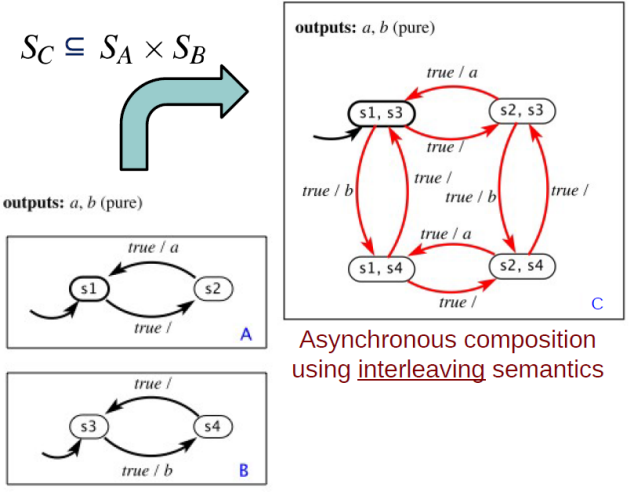
Synchronous composition

- In the equivalent model:
  - States = combination of states of the two FSMs
  - Transition = transition in FSM A and FSM B, happening simultaneously.
  - There might exist unreachable states in the equivalent model (states that will never be reached)

Asynchronous composition

Draw here

Asynchronous composition



Asynchronous composition using interleaving semantics

Note that now all states are reachable.

Figure 7: Asynchronous composition

Asynchronous composition

- In the equivalent model:
  - States = combination of states of the two FSMs
  - Transitions in the two FSMs can take place at irregular and independent (not synchronized) times
  - All states are reachable, in general
    - because one model can be much faster than the other

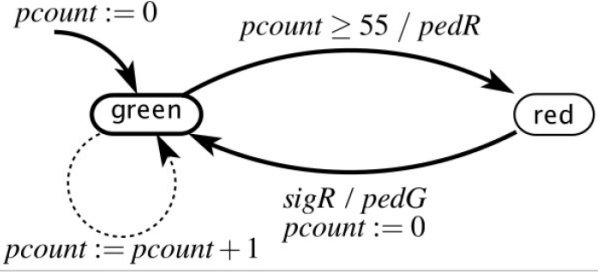
Asynchronous composition

Flavors of asynchronous composition

- How are simultaneous transitions handled?
- **Interleaving** semantics:
  - simultaneous transition in models A and B is not allowed (we may have either a transition in model A, or a transition in B)
  - i.e. transition in one model happens first, then transition in the other model B takes place (after some time delay)
- **Simultaneous** semantics:
  - simultaneous transition in models A and B is allowed
  - for example, we may have either
    - transition only in model A
    - transition only in model B
    - Simultaneous transition in models A and B

Example

variable: $pcount$: $\{0, \cdots, 55\}$
input: $sigR$: pure
outputs: $pedG$, $pedR$: pure



This light stays green for 55 seconds, then goes red.
Upon receiving a sigR input, it repeats the cycle.

Figure 8: Composition - Pedestrian Light

Example

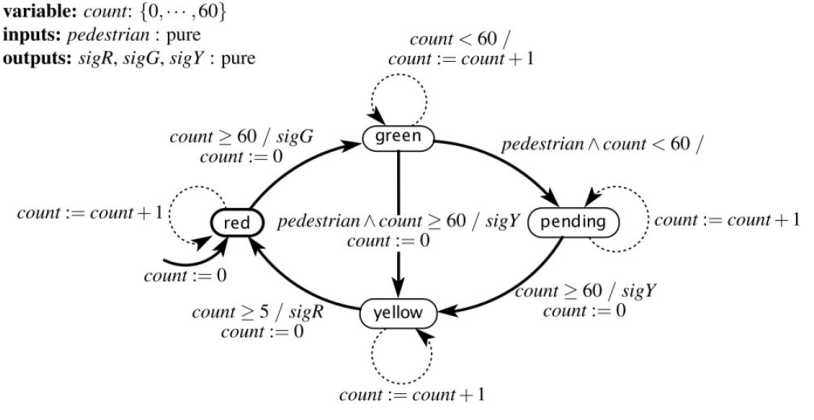


Figure 9: Composition - Car Light

Example

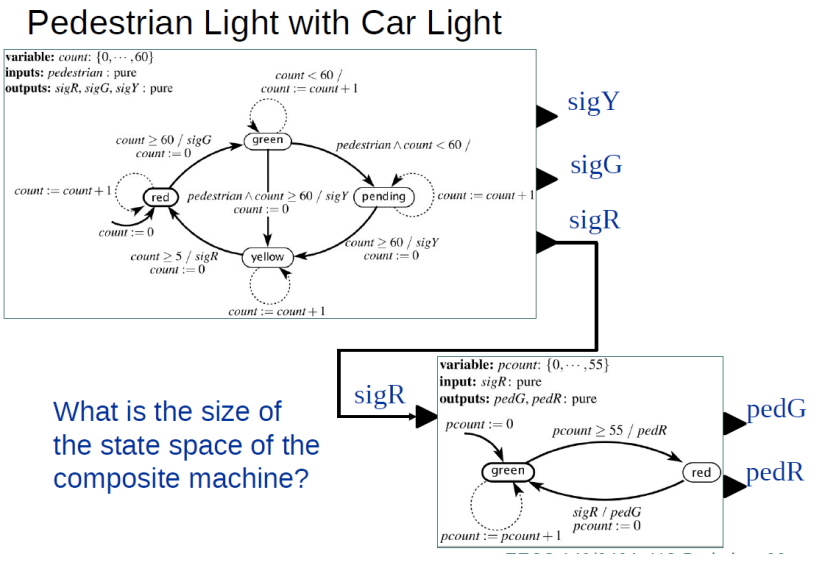
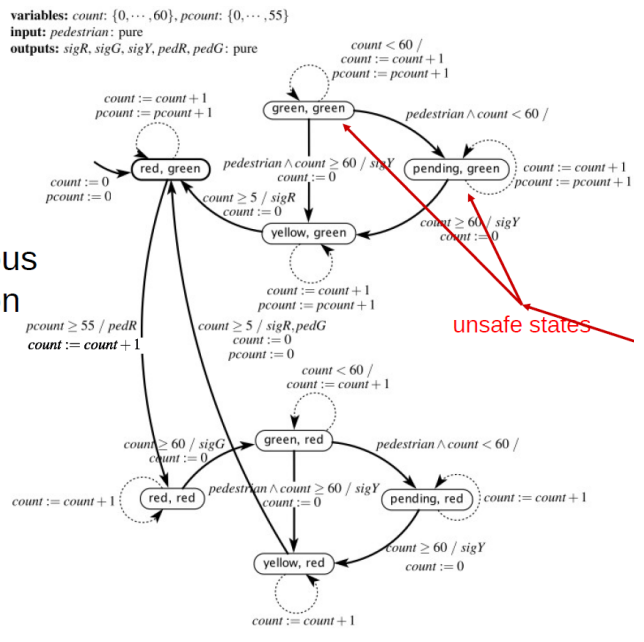


Figure 10: Cascade Composition - Both Lights

Example

Synchronous composition



Example

Synchronous composition with unreachable states removed

